Cybersecurity Sandbox Lab Syllabus

Hands-On Cybersecurity Lab Course Syllabus

Beginner level · No prior background required · KMUTT CPE

This course is fully hands-on. Every module is a lab: a machine to break into, a capture to analyse, or a defence to operate, THM/HTB style. Learners work from a provided Kali attacker delivered to the browser. Each lab checks out its target, or a pre-built multi-host bundle, from the image's template pool.

Sessions are stateless: when a session ends the attacker VM is destroyed and nothing on it survives. Targets, domains, logs, and sample data ship pre-seeded in the templates, so there is nothing to build. All flags and reports are submitted to the central portal.

Structure

- 10 core topics (65 labs) plus 1 optional topic (7 labs).
- 72 lab modules in total.
- Each lab runs 60–90 minutes; every topic ends with one unguided Challenge.

Table of Contents

Topic	Module	Lab	Objective	Tools
Topic 1 — Foundations of Security Practice				
1	1.1	Lab Environment Orientation & First Flag Capture	Log in to the Kali attacker in your browser and submit your first flag.	whoami, pwd, ls, date
1	1.2	Security Principles: CIA Triad & Ethical Conduct	Learn the three goals of security — confidentiality, integrity, and availability — plus the practical ethics of testing only what you are allowed to.	web browser, interactive quiz
1	1.3	Command-Line Fundamentals for Security Practitioners	Learn the basic Linux terminal commands and how to combine them.	man, --help, history, |, >
1	1.4	Terminal Text Editing & Script	Create, edit, and save files using the nano and vi text editors.	nano, vi, chmod +x, ./
1	1.5	Security Toolkit & Package Management	Find out which tools are installed and how to run them.	apt, which, dpkg -l, -h
1	1.6	Number Systems	Learn to read and convert binary and hexadecimal numbers.	python3, xxd, printf, bc
1	1.7	Internet Security Architecture	Learn how a web request travels from client to server using IP, ports, DNS, and HTTP.	dig, nc, curl -v, ping
Topic 2 — Linux Foundations for Cybersecurtiy				
2	2.1	Filesystem & shell hunt	Learn to move around the Linux filesystem and find files.	cd, ls, find, grep -r, locate, cat
2	2.2	Access Control: Users, Groups & File Permissions	Learn how Linux file permissions work and how to fix wrong ones.	chmod, chown, ls -l, sudo -l
2	2.3	Processes, services & cron	Learn to view and manage running processes, services, and scheduled jobs.	ps aux, top, kill, systemctl, crontab -l
2	2.4	Bash scripting	Write a simple Bash script to automate a repetitive task.	bash, grep, awk, sort, uniq -c
2	2.5	Remote access & transfer	Connect to another machine with SSH and copy files between them.	ssh, scp, tmux
2	2.6	Challenge: Bandit climb [Challenge]	Solve a multi-level Linux challenge using only the command line.	ssh, bash, find, grep, nc
Topic 3 — Network Traffic Analysis				
3	3.1	Network Layers & Ports	Learn the network layers and which ports common services use.	wireshark, tcpdump, nmap
3	3.2	Network Discovery	Work out the IP range of your network and find the live hosts on it.	ip a, ip r, ipcalc, nmap -sn

Topic	Module	Lab	Objective	Tools
3	3.3	Packet analysis	Open a packet capture and pull useful information out of it.	wireshark, tshark, follow tcp stream
3	3.4	Protocols hands-on	Talk to network services directly using DNS, HTTP, and raw connections.	dig, nslookup, curl -v, nc
3	3.5	Netcat toolkit	Use Netcat to transfer files, open listeners, and scan ports.	nc, ncat, nc -zv
3	3.6	Host firewalling	Write firewall rules on a Linux host to allow and block traffic.	ufw, iptables, nmap
3	3.7	Challenge: pcap forensics [Challenge]	Rebuild what happened in an attack using only a packet capture.	wireshark, tshark, strings
Topic 4 — Cryptography & Password Security				
4	4.1	Hashing, Salting & Keys	Learn what hashing, salting, and encryption keys actually do.	sha256sum, md5sum, openssl
4	4.2	Encoding vs Encryption vs Hashing	Tell apart encoding, encryption, and hashing, and reverse encoded data.	CyberChef, base64, xxd, tr
4	4.3	Password Cracking	Recover passwords from their hashes using a wordlist.	hashid, john, hashcat, rockyou.txt
4	4.4	Keys, signing & verification	Encrypt, sign, and verify files with openssl and gpg.	openssl enc, gpg --gen-key, gpg --verify
4	4.5	TLS inspection	Check a website's certificate and find weak encryption settings.	openssl s_client, nmap --script ssl-enum-ciphers, sslscan
4	4.6	Challenge: policy teardown [Challenge]	Crack a password dump and recommend a stronger password policy.	hashcat, john, pipal
Topic 5 — Reconnaissance & Enumeration				
5	5.1	Methodology & scan types	Learn the recon steps and what each type of Nmap scan does.	nmap -sS, nmap -sT, nmap -sU, nmap -sn
5	5.2	Passive Recon & OSINT	Collect information a target leaks through its documents and DNS records.	dnsrecon, theHarvester, exiftool, dig axfr
5	5.3	Service & version Enum	Identify exactly which services and versions a target runs.	nmap -sV, nmap -sC, enum4linux, snmpwalk
5	5.4	Vulnerability scanning	Run a scanner to find and rank a target's weaknesses.	openvas, nessus, searchsploit
5	5.5	Challenge: recon report [Challenge]	Produce a full reconnaissance report on an unknown target.	nmap, enum4linux, exiftool, searchsploit
Topic 6 — Web Application Security				
6	6.1	HTTP Request Anatomy	Learn how web requests and responses are structured.	Burp Suite, curl -v, browser devtools
6	6.2	Proxy & app mapping	Use a proxy to intercept requests and map a web app.	Burp Suite, ffuf, gobuster
6	6.3	Injection	Exploit inputs that aren't cleaned to reach the database or run commands.	sqlmap, Burp Suite, curl
6	6.4	Broken auth & sessions	Break weak login and session handling to access other accounts.	Burp Suite, jwt_tool, hydra
6	6.5	XSS & CSRF	Attack a web app's users with injected scripts and forged requests.	Burp Suite, XSS payloads, browser devtools
6	6.6	Access control & SSRF	Reach data and internal services you shouldn't be able to access.	Burp Suite, curl, ffuf
6	6.7	Challenge: Juice Shop hunt [Challenge]	Solve a wide set of web challenges on your own.	Burp Suite, sqlmap, ffuf
Topic 7 — Exploitation & Privilege Escalation				
7	7.1	Shells, payloads & enumeration	Learn how shells and payloads work before using them.	msfvenom, nc -lvnp, python3 -m http.server
7	7.2	Framework-driven exploitation	Use Metasploit to exploit a known vulnerability and get a shell.	msfconsole, meterpreter

Topic	Module	Lab	Objective	Tools
7	7.3	Manual exploitation	Exploit a target by hand using a public exploit.	searchsploit, gcc, python3, curl
7	7.4	Linux privilege escalation	Go from a normal user to root on a Linux machine.	linpeas, pspy, GTFOBins, find -perm
7	7.5	Windows privilege escalation	Go from a normal user to Administrator on a Windows machine.	winPEAS, PowerUp, accesschk
7	7.6	Post-exploitation	Collect credentials and set up access after breaking in.	mimikatz, lazagne, netstat
7	7.7	Challenge: root the box [Challenge]	Break into a machine and reach root on your own.	nmap, msfconsole, linpeas, winPEAS
Topic 8 — Network Attacks & Defense				
8	8.1	Man-in-the-middle	Intercept traffic between two machines on the same network.	arpspoof, ettercap, wireshark
8	8.2	Spoofing & capture	Capture login credentials by faking name-resolution replies.	Responder, wireshark, hashcat
8	8.3	DNS spoofing & SSL stripping	Redirect a victim's traffic and try to downgrade their HTTPS.	bettercap, dnsspoof, ettercap
8	8.4	Wireless capture analysis	Crack a Wi-Fi password from a captured handshake file.	aircrack-ng, hashcat, wireshark
8	8.5	Pivoting	Use a compromised machine to reach a hidden network behind it.	chisel, proxychains, ssh -L, nmap
8	8.6	Detection engineering	Write intrusion-detection rules that catch these network attacks.	snort, suricata, tcpreplay
8	8.7	Challenge: attack + detect [Challenge]	Run an attack, then write the detection that catches it.	ettercap, suricata, wireshark
Topic 9 — Defensive Security & Monitoring				
9	9.1	ATT&CK & detection concepts	Learn the MITRE ATT&CK framework and the basics of detection.	ATT&CK Navigator, grep
9	9.2	SIEM & log analysis	Search collected logs in a SIEM to find suspicious activity.	Splunk SPL, Kibana / KQL
9	9.3	Endpoint visibility	Use host logs to spot malicious activity on a Windows machine.	Sysmon, Get-WinEvent, Event Viewer
9	9.4	Threat hunting	Search telemetry for attacker activity that raised no alert.	Splunk, Zeek, jq
9	9.5	Incident triage	Handle the first hour of a security incident.	Splunk, Timesketch, IR playbook
9	9.6	Static malware triage	Examine a suspicious file without running it.	strings, pefile, floss, capa
9	9.7	Dynamic malware analysis	Run a malware sample in a sandbox and record what it does.	procmon, Regshot, Wireshark, Cuckoo
9	9.8	Challenge: investigate the breach [Challenge]	Investigate a breach across several data sources.	Splunk, Sysmon, Wireshark
Topic 10 — Adversary Simulation & Penetration Test Reporting				
10	10.1	Scope & recon	Agree the scope and gather information on the target range.	nmap, enum4linux, CherryTree
10	10.2	Breach & escalate	Break in and work up to full control of the range.	msfconsole, crackmapexec, mimikatz, chisel
10	10.3	Blue-team review	Find the traces your own attack left in the logs.	Splunk, Event Viewer, ATT&CK Navigator
10	10.4	Final report	Write a professional engagement report with recommended fixes.	report template, ATT&CK Navigator, CVSS calculator
10	10.5	Challenge: solo range [Challenge]	Break into a fresh range and report on it alone.	full toolkit
Optional — Windows & Active Directory *(optional)*				
Opt	O.1	Windows & AD concepts	Learn the basic concepts behind Windows and Active Directory.	PowerShell, nltest, wireshark

Topic	Module	Lab	Objective	Tools
Opt	O.2	Windows enumeration	Gather information about a Windows machine from the command line.	systeminfo, net user, Get-Process, ipconfig
Opt	O.3	Users, groups & NTFS	Learn how Windows controls access with users, groups, and permissions.	smbclient, smbmap, net view, icacls
Opt	O.4	Registry, services & tasks	Learn where Windows stores settings and how attackers hide persistence there.	reg query, sc query, schtasks, autoruns
Opt	O.5	Explore a live domain	Explore the users, groups, and policies in an Active Directory domain.	bloodhound, ldapsearch, Get-ADUser, gpresult
Opt	O.6	Windows event logs	Read Windows event logs to trace logons and program launches.	Event Viewer, Get-WinEvent, wevtutil
Opt	O.7	Challenge: domain audit [Challenge]	Audit a domain and list its security misconfigurations.	bloodhound, enum4linux, crackmapexec

Detailed Lab Breakdown

Topic 1 — Foundations of Security Practice

Start from nothing: learn the terminal, the tools, number systems, and the rules before any hacking.

1.1 Lab Environment Orientation & First Flag Capture

Objective: Log in to the Kali attacker in your browser and submit your first flag.
Environment: Kali attacker (Guacamole browser)
Lab tasks:

- Start the lab and wait for your Kali desktop to load in the browser
- Open a terminal and run whoami, pwd, ls, and date to see who and where you are
- Read the task brief and find the external flag-submission portal
- Submit the welcome flag to check the whole loop works

Tools / commands: whoami, pwd, ls, date
Submit to portal: *welcome flag*

1.2 Security Principles: CIA Triad & Ethical Conduct

Objective: Learn the three goals of security — confidentiality, integrity, and availability — plus the practical ethics of testing only what you are allowed to.
Environment: Kali attacker (browser) + interactive CIA & ethics quiz
Lab tasks:

- Learn what confidentiality, integrity, and availability each mean
- Read each scenario and decide which of the three goals it affects
- Learn the practical ethics: authorization, staying in scope, and responsible disclosure
- Answer the quiz questions and check your score

Tools / commands: web browser, interactive quiz
Submit to portal: *your quiz score (all correct)*

1.3 Command-Line Fundamentals for Security Practitioners

Objective: Learn the basic Linux terminal commands and how to combine them.
Environment: Kali attacker (browser)
Lab tasks:

- Run commands with options, and read their help with man and --help
- Save typing with tab-completion and the up-arrow command history
- Send one command's output into another using a pipe (|)
- Save a command's output to a file using redirection (>)

Tools / commands: man, --help, history, |, >
Submit to portal: *output of the required recipe*

1.4 Terminal Text Editing & Script

Objective: Create, edit, and save files using the nano and vi text editors.
Environment: Kali attacker (browser)
Lab tasks:

- Open a file in nano, change it, save, and exit
- Do the same in vi: enter insert mode, type, then save and quit
- Create a small text file and a short shell script
- Make the script runnable with chmod +x and run it

Tools / commands: nano, vi, chmod +x, ./
Submit to portal: *the string your script prints*

1.5 Security Toolkit & Package Management

Objective: Find out which tools are installed and how to run them.
Environment: Kali attacker (browser; egress if the lab allows)
Lab tasks:

- List installed tools and read one tool's help with -h
- Search for a package with apt search

- Learn what 'apt update' and 'apt upgrade' do
- Run one tool against a sample file to see its output

Tools / commands: apt, which, dpkg -l, -h
Submit to portal: *version string of the required tool*

1.6 Number Systems

Objective: Learn to read and convert binary and hexadecimal numbers.
Environment: Kali attacker (browser)
Lab tasks:

- Count in binary and hex, and convert between them and decimal
- See why one part of an IP address (an octet) only goes up to 255
- View a file's raw bytes with xxd and match them to ASCII letters
- Convert the given values using python3 or printf

Tools / commands: python3, xxd, printf, bc
Submit to portal: *the decoded hex value*

1.7 How the internet works

Objective: Learn how a web request travels from client to server using IP, ports, DNS, and HTTP.
Environment: Kali + provided target
Lab tasks:

- Turn a website name into an IP address with dig
- Connect to a port with nc and read the service's banner
- Send a web request with curl -v and read the reply
- Point out the request line, the headers, and the status code

Tools / commands: dig, nc, curl -v, ping
Submit to portal: *the HTTP status + server header*

Topic 2 — Linux Foundations for Cybersecurtiy

Get comfortable using Linux from the command line.

2.1 Filesystem & shell hunt

Objective: Learn to move around the Linux filesystem and find files.
Environment: Kali + provided flag-seeded Linux target
Lab tasks:

- Move between folders with cd, ls, and pwd
- Read files with cat, less, head, and tail
- Search inside files with grep -r, and find files with find and locate
- Combine commands with pipes to narrow your search

Tools / commands: cd, ls, find, grep -r, locate, cat
Submit to portal: *5 flags hidden under /opt and /var*

2.2 Access Control: Users, Groups & File Permissions

Objective: Learn how Linux file permissions work and how to fix wrong ones.
Environment: Kali + provided multi-user Linux target
Lab tasks:

- Read a file's permissions (rwx) and owner with ls -l
- Change permissions and owner with chmod and chown to fix a broken website folder
- Check what you are allowed to run as admin with sudo -l
- Find the file that everyone can write to but shouldn't

Tools / commands: chmod, chown, ls -l, sudo -l
Submit to portal: *the misconfigured file path*

2.3 Processes, services & cron

Objective: Learn to view and manage running processes, services, and scheduled jobs.
Environment: Kali + provided Linux target (rogue job planted)
Lab tasks:

- List running programs with ps and top
- Stop a program with kill
- Start, stop, and check background services with systemctl
- List scheduled jobs with crontab -l and find the suspicious one

Tools / commands: ps aux, top, kill, systemctl, crontab -l
Submit to portal: *the rogue cron line*

2.4 Bash scripting

Objective: Write a simple Bash script to automate a repetitive task.
Environment: Kali (sample auth.log shipped on the box)
Lab tasks:

- Use variables, loops, and if-statements in a Bash script
- Write a script that reads a log and pulls out failed-login IP addresses
- Count how often each IP appears with sort and uniq -c
- Print the ten most frequent IP addresses

Tools / commands: bash, grep, awk, sort, uniq -c
Submit to portal: *the top offending IP*

2.5 Remote access & transfer

Objective: Connect to another machine with SSH and copy files between them.
Environment: Kali + provided Linux target
Lab tasks:

- Log in to the target over SSH from Kali
- Copy a file to and from the target with scp
- Keep your session open with basic tmux commands
- Download a file from the target and read it on Kali

Tools / commands: ssh, scp, tmux
Submit to portal: *contents of the transferred file*

2.6 Challenge: Bandit climb [Challenge]

Objective: Solve a multi-level Linux challenge using only the command line.
Environment: Provided 12-level Linux target (OverTheWire-style)
Lab tasks:

- Work through each level using only terminal commands
- Use each level's password to unlock the next one
- Solve it without looking up a walkthrough

Tools / commands: ssh, bash, find, grep, nc
Deliverable: *final-level password*

Topic 3 — Network Traffic Analysis

Learn how networks work: addresses, packets, protocols, and firewalls.

3.1 Network Layers & Ports

Objective: Learn the network layers and which ports common services use.
Environment: Kali + provided pcap + target
Lab tasks:

- Match parts of a captured packet to the network layers (OSI / TCP-IP)
- Match common port numbers to their services (80 = HTTP, 53 = DNS, …)
- Spot ARP and DHCP messages in the capture and learn what they do
- Open the same traffic in both tcpdump and Wireshark

Tools / commands: wireshark, tcpdump, nmap
Submit to portal: *service on the unknown port*

3.2 Network Discovery

Objective: Work out the IP range of your network and find the live hosts on it.
Environment: Kali + provided target on your lab segment

Lab tasks:

- Check your IP and routes with ip a and ip r
- Work out which IP addresses belong to the network
- Find which hosts are switched on with nmap -sn
- Confirm which IP address is your target

Tools / commands: ip a, ip r, ipcalc, nmap -sn
Submit to portal: *segment CIDR + target IP*

3.3 Packet analysis

Objective: Open a packet capture and pull useful information out of it.
Environment: Kali + provided pcap set (shipped on the box)
Lab tasks:

- Filter the capture to show only HTTP, DNS, or a chosen port
- Right-click a packet and 'Follow TCP stream' to read a full conversation
- Find a username and password sent in clear text
- Save a file that was transferred inside the capture

Tools / commands: wireshark, tshark, follow tcp stream
Submit to portal: *the recovered credential pair*

3.4 Protocols hands-on

Objective: Talk to network services directly using DNS, HTTP, and raw connections.
Environment: Kali + provided DNS/HTTP target
Lab tasks:

- Look up DNS records with dig and nslookup
- Send a web request by hand with curl -v
- Connect straight to a port with nc and type to the service
- Compare how TCP and UDP behave (optional)

Tools / commands: dig, nslookup, curl -v, nc
Submit to portal: *TXT record value from the target zone*

3.5 Netcat toolkit

Objective: Use Netcat to transfer files, open listeners, and scan ports.
Environment: Kali + provided target
Lab tasks:

- Send a file to and from the target using nc
- Connect to a listener (a waiting port) running on the target
- Receive a connection the target sends back to your Kali
- Check which ports are open with nc -zv

Tools / commands: nc, ncat, nc -zv
Submit to portal: *banner from the open service*

3.6 Host firewalling

Objective: Write firewall rules on a Linux host to allow and block traffic.
Environment: Kali + provided Linux target you administer
Lab tasks:

- Write allow and block rules with ufw or iptables on the target
- Block everything incoming by default, then allow one service
- Confirm the result by scanning the target from Kali
- Turn on logging for blocked packets

Tools / commands: ufw, iptables, nmap
Submit to portal: *the one reachable port after lockdown*

3.7 Challenge: pcap forensics [Challenge]

Objective: Rebuild what happened in an attack using only a packet capture.
Environment: Kali + provided mixed pcap
Lab tasks:

- Put the attacker's actions in order to build a timeline
- Identify the attacker's IP, the target, and how they got in
- Recover any files the attacker stole

Tools / commands: wireshark, tshark, strings
Deliverable: *attacker IP + exfil filename*

Topic 4 — Cryptography & Password Security

Learn the difference between encoding, encryption, and hashing, then crack weak passwords.

4.1 Hashing, Salting & Keys

Objective: Learn what hashing, salting, and encryption keys actually do.
Environment: Kali attacker (browser)
Lab tasks:

- Hash the same text twice and see the result is identical
- Add a salt and watch the hash change
- Encrypt one file with a shared password, and one with a key pair
- Say when you would use each method

Tools / commands: sha256sum, md5sum, openssl
Submit to portal: *the two hashes you produced*

4.2 Encoding vs Encryption vs Hashing

Objective: Tell apart encoding, encryption, and hashing, and reverse encoded data.
Environment: Kali + CyberChef (local) + provided strings file
Lab tasks:

- Recognise Base64, hex, URL, and ROT encodings
- Decode a value that has been encoded several times over
- Learn why encoding does not protect data
- Tell a hash apart from encrypted text

Tools / commands: CyberChef, base64, xxd, tr
Submit to portal: *the plaintext at the end of the chain*

4.3 Password Cracking

Objective: Recover passwords from their hashes using a wordlist.
Environment: Kali + provided hash set + rockyou (CPU/GPU quota)
Lab tasks:

- Identify the hash type with hashid
- Crack the hashes with a wordlist using john or hashcat
- Improve results with rules and masks
- Crack a set of salted hashes

Tools / commands: hashid, john, hashcat, rockyou.txt
Submit to portal: *3 cracked passwords*

4.4 Keys, signing & verification

Objective: Encrypt, sign, and verify files with openssl and gpg.
Environment: Kali + openssl / gpg + provided files
Lab tasks:

- Encrypt and decrypt a file with a password (AES) using openssl
- Create a public/private key pair with gpg
- Sign a file and verify the signature
- Catch a changed file when the signature fails to verify

Tools / commands: openssl enc, gpg --gen-key, gpg --verify
Submit to portal: *signer's key fingerprint*

4.5 TLS inspection

Objective: Check a website's certificate and find weak encryption settings.
Environment: Kali + provided HTTPS target (weak config)

Lab tasks:

- View the site's certificate with openssl s_client
- Check its expiry date, chain, and listed names (SAN)
- List the encryption options with nmap's ssl-enum script
- Point out the outdated protocol or weak cipher

Tools / commands: openssl s_client, nmap --script ssl-enum-ciphers, sslscan
Submit to portal: *the deprecated protocol in use*

4.6 Challenge: policy teardown [Challenge]

Objective: Crack a password dump and recommend a stronger password policy.
Environment: Kali + provided leaked-hash dump
Lab tasks:

- Crack the provided set of password hashes
- Look for reused passwords and common patterns
- Recommend a clear password and MFA policy

Tools / commands: hashcat, john, pipal
Deliverable: *crack rate % + top weakness*

Topic 5 — Reconnaissance & Enumeration

Gather information about a target and scan it for weaknesses before attacking.

5.1 Methodology & scan types

Objective: Learn the recon steps and what each type of Nmap scan does.
Environment: Kali + provided target
Lab tasks:

- Walk through the recon steps: passive, then active, then detailed enumeration
- Run and compare the nmap scans -sS, -sT, -sU, and -sn
- Learn what each scan actually sends and when to use it
- Write your results into your notes

Tools / commands: nmap -sS, nmap -sT, nmap -sU, nmap -sn
Submit to portal: *open ports from the SYN scan*

5.2 Passive Recon & OSINT

Objective: Collect information a target leaks through its documents and DNS records.
Environment: Kali + provided intranet-mirror target (docs, metadata, DNS zone)
Lab tasks:

- Read the target's DNS records and list its subdomains
- Find emails and usernames inside the hosted documents
- Read document metadata for software names and authors
- Build a one-page profile of the target

Tools / commands: dnsrecon, theHarvester, exiftool, dig axfr
Submit to portal: *count of discovered subdomains*

5.3 Service & version enum

Objective: Identify exactly which services and versions a target runs.
Environment: Kali + provided multi-service target
Lab tasks:

- Detect service versions with nmap -sV
- Run nmap's default scripts for extra detail
- Dig into SMB, FTP, or SNMP services where present
- Note each version to look up known vulnerabilities offline

Tools / commands: nmap -sV, nmap -sC, enum4linux, snmpwalk
Submit to portal: *version of the web server*

5.4 Vulnerability scanning

Objective: Run a scanner to find and rank a target's weaknesses.

Environment: Kali + pre-synced OpenVAS/Nessus + provided vulnerable target
Lab tasks:

- Run the scanner once without credentials and once with them
- Read each finding's CVE id and CVSS score
- Separate genuine issues from false alarms
- Export the findings sorted by priority

Tools / commands: openvas, nessus, searchsploit
Submit to portal: *highest-CVSS finding*

5.5 Challenge: recon report [Challenge]

Objective: Produce a full reconnaissance report on an unknown target.
Environment: Kali + provided unknown target bundle
Lab tasks:

- Do full passive and active recon offline
- List every host, its services, and its likely weaknesses
- Deliver a clear findings report

Tools / commands: nmap, enum4linux, exiftool, searchsploit
Deliverable: *ranked attack-surface summary*

Topic 6 — Web Application Security

Learn the most common web app vulnerabilities, one broken app at a time (OWASP Top 10).

6.1 HTTP Request Anatomy

Objective: Learn how web requests and responses are structured.
Environment: Kali + Burp + provided web app
Lab tasks:

- Look at request methods, status codes, and headers
- Watch a cookie get set and then sent back
- Change a request and send it again
- Trace one full request-and-response cycle

Tools / commands: Burp Suite, curl -v, browser devtools
Submit to portal: *the session cookie name*

6.2 Proxy & app mapping

Objective: Use a proxy to intercept requests and map a web app.
Environment: Kali + Burp + provided DVWA container
Lab tasks:

- Set up the Burp proxy and install its certificate
- Pause a request, edit it, and let it continue
- Build a map of the app's pages in Burp
- Find hidden pages by brute-forcing paths with ffuf or gobuster

Tools / commands: Burp Suite, ffuf, gobuster
Submit to portal: *a hidden endpoint you found*

6.3 Injection

Objective: Exploit inputs that aren't cleaned to reach the database or run commands.
Environment: Kali + provided DVWA / bWAPP container
Lab tasks:

- Test input fields for SQL injection by hand
- Pull data out of the database (UNION or error-based)
- Confirm you can run system commands through an input
- Note the fix: use parameterized queries

Tools / commands: sqlmap, Burp Suite, curl
Submit to portal: *admin hash from the DB*

6.4 Broken auth & sessions

Objective: Break weak login and session handling to access other accounts.
Environment: Kali + provided Juice Shop container
Lab tasks:

- Try common passwords and check whether lockout is missing
- Inspect and tamper with the session token
- Test the password-reset process for flaws
- Gain access to another user's account

Tools / commands: Burp Suite, jwt_tool, hydra
Submit to portal: *another user's session*

6.5 XSS & CSRF

Objective: Attack a web app's users with injected scripts and forged requests.
Environment: Kali + provided DVWA + Juice Shop containers
Lab tasks:

- Trigger reflected, stored, and DOM cross-site scripting
- Write a script that steals a cookie (inside the lab only)
- Build a proof-of-concept CSRF request
- Recommend the fixes: output encoding and anti-CSRF tokens

Tools / commands: Burp Suite, XSS payloads, browser devtools
Submit to portal: *stored-XSS trigger location*

6.6 Access control & SSRF

Objective: Reach data and internal services you shouldn't be able to access.
Environment: Kali + provided vulnerable web container
Lab tasks:

- Change an id in the URL to read someone else's data (IDOR)
- Upload a file that the app should have rejected
- Make the server fetch an internal address for you (SSRF)
- Check whether security headers are set

Tools / commands: Burp Suite, curl, ffuf
Submit to portal: *internal service reached via SSRF*

6.7 Challenge: Juice Shop hunt [Challenge]

Objective: Solve a wide set of web challenges on your own.
Environment: Kali + provided full Juice Shop container
Lab tasks:

- Complete a set of Juice Shop challenges from the score board
- Cover at least three different vulnerability types
- Write down how you solved each one

Tools / commands: Burp Suite, sqlmap, ffuf
Deliverable: *score-board completion set*

Topic 7 — Exploitation & Privilege Escalation

Break into a machine and work up to full administrator control, on Linux and Windows.

7.1 Shells, payloads & enumeration

Objective: Learn how shells and payloads work before using them.
Environment: Kali + provided target
Lab tasks:

- Learn the difference between a bind shell and a reverse shell
- Build a payload with msfvenom
- Send it to the target and catch the connection with nc
- Always gather information before trying an exploit

Tools / commands: msfvenom, nc -lvnp, python3 -m http.server
Submit to portal: *hostname from your first shell*

7.2 Framework-driven exploitation

Objective: Use Metasploit to exploit a known vulnerability and get a shell.
Environment: Kali + Metasploit + provided vulnerable target
Lab tasks:

- Search Metasploit for a module that matches the target
- Set the required options and choose a payload
- Run it to get a session on the target
- Make the shell stable and upgrade it

Tools / commands: msfconsole, meterpreter
Submit to portal: *user flag on the target*

7.3 Manual exploitation

Objective: Exploit a target by hand using a public exploit.
Environment: Kali + provided vulnhub-style target
Lab tasks:

- Match a public exploit to the service's version
- Adjust the exploit code for this target
- Run it to get command execution
- Make sure you understand each step

Tools / commands: searchsploit, gcc, python3, curl
Submit to portal: *shell as the service user*

7.4 Linux privilege escalation

Objective: Go from a normal user to root on a Linux machine.
Environment: Kali + provided misconfigured Linux target
Lab tasks:

- Look for weaknesses with linpeas and by hand
- Abuse a SUID file, a sudo rule, or a cron job
- Use a writable path or weak service to gain root
- Read the root flag

Tools / commands: linpeas, pspy, GTFOBins, find -perm
Submit to portal: *root flag*

7.5 Windows privilege escalation

Objective: Go from a normal user to Administrator on a Windows machine.
Environment: Kali + provided misconfigured Windows target
Lab tasks:

- Look for weaknesses with winPEAS
- Abuse an unquoted service path or weak permissions
- Use a token or vulnerable service to become Administrator
- Read the administrator flag

Tools / commands: winPEAS, PowerUp, accesschk
Submit to portal: *administrator flag*

7.6 Post-exploitation

Objective: Collect credentials and set up access after breaking in.
Environment: Kali + provided foothold target
Lab tasks:

- Gather more information now that you have access
- Collect saved passwords and credentials from the host
- Set up a way to keep access (inside the lab only)
- Write down the impact in plain terms

Tools / commands: mimikatz, lazagne, netstat
Submit to portal: *second set of credentials found*

7.7 Challenge: root the box **[Challenge]**

Objective: Break into a machine and reach root on your own.
Environment: Kali + provided fresh target
Lab tasks:

- Go from recon to a foothold to root on your own
- Capture both the user and root flags
- Write up the steps you took

Tools / commands: nmap, msfconsole, linpeas, winPEAS
Deliverable: *user + root flags*

Topic 8 — Network Attacks & Defense

Attack a ready-made network, then write the detection rules that would catch the attack.

8.1 Man-in-the-middle

Objective: Intercept traffic between two machines on the same network.
Environment: Kali + provided multi-host bundle (two victim hosts)
Lab tasks:

- Trick two hosts into sending traffic through you (ARP spoofing)
- Place your Kali in the middle of their conversation
- Capture the traffic now passing through you
- Understand why a switch alone does not prevent this

Tools / commands: arpspoof, ettercap, wireshark
Submit to portal: *captured credential on the wire*

8.2 Spoofing & capture

Objective: Capture login credentials by faking name-resolution replies.
Environment: Kali + provided bundle with a victim host
Lab tasks:

- Run Responder to answer name-lookup requests falsely
- Capture the login data victims send (inside the lab only)
- Explain the defenses: network segmentation and signing

Tools / commands: Responder, wireshark, hashcat
Submit to portal: *captured hash*

8.3 DNS spoofing & SSL stripping

Objective: Redirect a victim's traffic and try to downgrade their HTTPS.
Environment: Kali + provided multi-host bundle
Lab tasks:

- Give the victim fake DNS answers to send them elsewhere
- Try to downgrade HTTPS links to plain HTTP
- See where HSTS stops the downgrade
- Recommend the defenses that prevent it

Tools / commands: bettercap, dnsspoof, ettercap
Submit to portal: *the spoofed domain's planted record*

8.4 Wireless capture analysis

Objective: Crack a Wi-Fi password from a captured handshake file.
Environment: Kali + provided WPA2 handshake capture file
Lab tasks:

- Open the captured Wi-Fi handshake in Wireshark or aircrack
- Crack a weak Wi-Fi password offline with hashcat
- Learn why the WPA2 handshake can be attacked
- Recommend WPA3 and a strong passphrase

Tools / commands: aircrack-ng, hashcat, wireshark
Submit to portal: *recovered test passphrase*

8.5 Pivoting

Objective: Use a compromised machine to reach a hidden network behind it.
Environment: Kali + provided dual-homed bundle (foothold + hidden host)
Lab tasks:

- Take control of the host that sits on two networks
- Set up a tunnel through it with chisel or proxychains
- Scan the hidden network through that tunnel
- Reach a service on the inner network

Tools / commands: chisel, proxychains, ssh -L, nmap
Submit to portal: *service on the inner subnet*

8.6 Detection engineering

Objective: Write intrusion-detection rules that catch these network attacks.
Environment: Kali + provided IDS host (Snort/Suricata pre-installed)
Lab tasks:

- Review how the provided IDS is set up
- Write a rule that detects ARP spoofing
- Write a rule that detects the pivot scan
- Test the rules by replaying a captured attack

Tools / commands: snort, suricata, tcpreplay
Submit to portal: *alert count on replay*

8.7 Challenge: attack + detect [Challenge]

Objective: Run an attack, then write the detection that catches it.
Environment: Kali + provided network bundle with IDS
Lab tasks:

- Carry out one network attack
- Write a detection rule that fires on it
- Prove it works with before-and-after logs

Tools / commands: ettercap, suricata, wireshark
Deliverable: *working rule + firing alert*

Topic 9 — Defensive Security & Monitoring

Work as a defender: use logs and tools to spot, investigate, and report an attack.

9.1 ATT&CK & detection concepts

Objective: Learn the MITRE ATT&CK framework and the basics of detection.
Environment: Kali + provided offline ATT&CK reference + sample events
Lab tasks:

- Browse the MITRE ATT&CK tactics and techniques
- Match sample events to the techniques they show
- Learn the difference between an IOC and an IOA
- Note which log source reveals which technique

Tools / commands: ATT&CK Navigator, grep
Submit to portal: *the technique ID behind the sample*

9.2 SIEM & log analysis

Objective: Search collected logs in a SIEM to find suspicious activity.
Environment: Kali browser + provided SIEM target (Splunk/ELK, data pre-loaded)
Lab tasks:

- Open the SIEM web page from Kali
- Write searches to find failed logins
- Add a panel to a dashboard
- Create a simple alert

Tools / commands: Splunk SPL, Kibana / KQL
Submit to portal: *count of brute-force attempts*

9.3 Endpoint visibility

Objective: Use host logs to spot malicious activity on a Windows machine.
Environment: Kali + provided Windows target (Sysmon pre-deployed, events seeded)
Lab tasks:

- Look at how Sysmon is configured
- Find process-start and network-connection events
- Separate normal activity from suspicious activity
- Link the events to known attacker behaviour

Tools / commands: Sysmon, Get-WinEvent, Event Viewer
Submit to portal: *the malicious parent-child process*

9.4 Threat hunting

Objective: Search telemetry for attacker activity that raised no alert.
Environment: Kali + provided telemetry dataset target
Lab tasks:

- Start from a guess based on ATT&CK (a hypothesis)
- Search the data for signs of that technique
- Confirm or rule it out
- Write up what you checked and found

Tools / commands: Splunk, Zeek, jq
Submit to portal: *the technique ID you confirmed*

9.5 Incident triage

Objective: Handle the first hour of a security incident.
Environment: Kali + provided alert-queue target
Lab tasks:

- Judge how serious the alert is
- Collect evidence from memory and the host
- Decide whether to contain it or keep watching
- Build a timeline of the incident

Tools / commands: Splunk, Timesketch, IR playbook
Submit to portal: *patient-zero host*

9.6 Static malware triage

Objective: Examine a suspicious file without running it.
Environment: Provided isolated analysis VM + lab samples (no egress, snapshot lifecycle)
Lab tasks:

- Hash the file and check it offline against known-bad lists
- Pull readable text and imported functions from the file
- List the indicators of compromise you find
- Write a short triage note

Tools / commands: strings, pefile, floss, capa
Submit to portal: *3 extracted IOCs*

9.7 Dynamic malware analysis

Objective: Run a malware sample in a sandbox and record what it does.
Environment: Provided isolated analysis VM + sandbox (no egress, snapshot lifecycle)
Lab tasks:

- Run the sample inside the isolated sandbox
- Record what files, registry keys, and network traffic it creates
- Compare this with what the static analysis showed
- Restore the clean snapshot when finished

Tools / commands: procmon, Regshot, Wireshark, Cuckoo
Submit to portal: *the sample's network beacon target*

9.8 Challenge: investigate the breach **[Challenge]**

Objective: Investigate a breach across several data sources.
Environment: Kali + provided multi-source analysis bundle
Lab tasks:

- Combine SIEM, endpoint, and network data
- Rebuild the full chain of the attack
- Deliver an incident report listing the indicators of compromise

Tools / commands: Splunk, Sysmon, Wireshark
Deliverable: *timeline + IOC list*

Topic 10 — Adversary Simulation & Penetration Test Reporting

Put everything together on one ready-made range: break in, then show you can detect it.

10.1 Scope & recon

Objective: Agree the scope and gather information on the target range.
Environment: Kali + provided capstone range bundle
Lab tasks:

- Note the agreed scope and rules
- Do full recon on the range (offline)
- Decide which targets to hit first by likely impact
- Record everything for the report

Tools / commands: nmap, enum4linux, CherryTree
Submit to portal: *ranked target list*

10.2 Breach & escalate

Objective: Break in and work up to full control of the range.
Environment: Kali + provided range bundle (AD + Linux + web)
Lab tasks:

- Get an initial foothold on the range
- Raise your privileges on that machine
- Move from one machine to the next
- Reach the domain controller or main target

Tools / commands: msfconsole, crackmapexec, mimikatz, chisel
Submit to portal: *domain admin / final flag*

10.3 Blue-team review

Objective: Find the traces your own attack left in the logs.
Environment: Kali + range SIEM/logs (shipped with the bundle)
Lab tasks:

- Find your own activity in the logs
- Note what was detected and what was missed
- Suggest detections for the gaps
- Say what you would change as a defender

Tools / commands: Splunk, Event Viewer, ATT&CK Navigator
Submit to portal: *detection-gap list*

10.4 Final report

Objective: Write a professional engagement report with recommended fixes.
Environment: Report template + ATT&CK matrix (submitted to portal)
Lab tasks:

- Write a short summary and risk ratings for managers
- Write the step-by-step technical details
- Link each step to a MITRE ATT&CK technique
- List clear, prioritised fixes

Tools / commands: report template, ATT&CK Navigator, CVSS calculator
Submit to portal: *complete engagement report*

10.5 Challenge: solo range [Challenge]

Objective: Break into a fresh range and report on it alone.
Environment: Kali + provided unseen capstone range bundle
Lab tasks:

- Break into a brand-new range on your own, against the clock
- Capture every flag
- Submit a full report before the deadline

Tools / commands: full toolkit
Deliverable: *all flags + report, solo*

Optional — Windows & Active Directory

> **OPTIONAL MODULE — may be skipped or taken after the core path. Included for learners heading toward enterprise / Windows environments.**

Explore and audit a ready-made Windows domain — you test it, you don't set it up.

O.1 Windows & AD concepts

Objective: Learn the basic concepts behind Windows and Active Directory.
Environment: Kali + provided Windows target / AD bundle
Lab tasks:

- Learn the difference between a workgroup and a domain, and why companies use Active Directory
- See NTLM and Kerberos logins happen in a capture or log
- Learn PowerShell basics: commands (cmdlets) and the pipeline
- Identify which machine is the domain controller

Tools / commands: PowerShell, nltest, wireshark
Submit to portal: *the domain's name + DC hostname*

O.2 Windows enumeration

Objective: Gather information about a Windows machine from the command line.
Environment: Kali + provided Windows target
Lab tasks:

- Connect to the Windows target (RDP or console) through the browser
- Move around using Command Prompt and PowerShell
- Check system details, user accounts, and network settings
- List installed software and which updates are missing

Tools / commands: systeminfo, net user, Get-Process, ipconfig
Submit to portal: *OS build + logged-in user*

O.3 Users, groups & NTFS

Objective: Learn how Windows controls access with users, groups, and permissions.
Environment: Kali + provided Windows target (leaky share)
Lab tasks:

- List the local users and groups
- Compare NTFS permissions with share permissions
- List shared folders with smbclient or net view
- Find the share that gives too much access

Tools / commands: smbclient, smbmap, net view, icacls
Submit to portal: *file readable via the weak share*

O.4 Registry, services & tasks

Objective: Learn where Windows stores settings and how attackers hide persistence there.
Environment: Kali + provided Windows target (persistence planted)
Lab tasks:

- Browse the Windows registry
- List services and the program each one runs
- List scheduled tasks
- Find where the attacker set something to run automatically

Tools / commands: reg query, sc query, schtasks, autoruns
Submit to portal: *the persistence autorun entry*

O.5 Explore a live domain

Objective: Explore the users, groups, and policies in an Active Directory domain.
Environment: Kali + provided AD bundle (DC + client, pre-built)
Lab tasks:

- List the domain's users, groups, and containers (OUs)
- Map who belongs to which group using BloodHound
- Review the Group Policy settings in effect
- Identify the high-privilege (admin) accounts

Tools / commands: bloodhound, ldapsearch, Get-ADUser, gpresult
Submit to portal: *domain FQDN + a Domain Admin account*

O.6 Windows event logs

Objective: Read Windows event logs to trace logons and program launches.
Environment: Kali + provided AD bundle (logs pre-seeded)
Lab tasks:

- Open the logs in Event Viewer or with Get-WinEvent
- Find a successful login (event 4624) and a failed one (4625)
- Find a program-start event (4688)
- Match a run of failed logins followed by a success

Tools / commands: Event Viewer, Get-WinEvent, wevtutil
Submit to portal: *account tied to the brute-force*

O.7 Challenge: domain audit [Challenge]

Objective: Audit a domain and list its security misconfigurations.
Environment: Kali + provided misconfigured AD bundle
Lab tasks:

- List the domain's users, groups, shares, and policies
- Find at least five misconfigurations
- Rank them by how serious they are in a short report

Tools / commands: bloodhound, enum4linux, crackmapexec
Deliverable: *5 findings, ranked*